

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 4: BOUNDARY TRACKING PROCEDURES

SLO 2: Analyze and represent object shapes using boundary coordinates for further shape-based analysis.

COORDINATE EXPLORATION SOLUTIONS

Binary Grid (Row, Column Indexing):

	0	1	2	3	4
0	0	0	0	0	0
1	0	1	1	1	0
2	0	1	1	1	0
3	0	1	1	1	0
4	0	0	0	0	0

1. Coordinates of four boundary pixels (Row, Column format):

(1, 1), (1, 2), (1, 3), (2, 3)

Note: Any 4 of the outer edge pixels—(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)—are correct boundary pixels.

2. Interior pixel of the object:

(2, 2)

PATTERN RECOGNITION SOLUTIONS

Given Coordinates:

- **Object A:** (1,1), (1,2), (1,3), (2,3), (3,3), (3,2), (3,1), (2,1)
- **Object B:** (1,2), (2,3), (3,2), (2,1)

1. Which object likely represents a square shape?

Object A (Forms an axis-aligned 3x3 outer boundary perimeter).

2. Which object likely represents a diamond or rotated shape?

Object B (Connects four diagonal midpoints forming a rotated square/diamond).

3. Explain how boundary coordinates help identify shapes:

Boundary coordinates provide ordered geometric points that define spatial structure. By calculating distance vectors, directional changes (chain codes), curvature, perimeter length, and aspect ratios between sequential boundary points, algorithms can extract rotation and scale-invariant descriptors to accurately classify objects.